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1. BLAST can be used for both protein and nucleotide sequences. (True/False) [1]
2. Time complexity of multiple sequence alignment using DP for t sequences of length n : [1]
 - (a) $O(n^2)$
 - (b) $O(n^3)$
 - (c) $O(n^t 2^t)$
 - (d) $O(n^t \cdot 2^n)$
3. Which is NOT a limitation of global alignment? [1]
 - (a) Cannot handle variable fragments like introns
 - (b) Cannot detect local similarities
 - (c) Cannot align two sequences of same length
 - (d) Misaligns conserved regions
4. In affine gap penalties, if opening penalty=-5 and extension=-1, the cost of a gap of length 4 is: [1]
 - (a) -5
 - (b) -8
 - (c) -9
 - (d) -2
5. Give the alignment matrix of the sequences 'AATCGCGCGGT' and 'ATGCGCCGT' assuming the following costs: $Cost(a, a) = 0$, $Cost(a, b) = 3$, when $a \neq b$, $Cost(a, -) = Cost(-, a) = 2$. Also show the alignment. [4+2=6]